

Gabarito

- ① Temos que $X =$ "Número de motoristas que param dos 10 selecionados". Sabemos que $X \sim \text{Bin}(10; 0,35)$.

1.1) Desejamos encontrar

$$P[X=5] = \binom{10}{5} 0,35^5 0,65^5 = \frac{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5!}{5! \cdot 5!} 0,00060941$$

$$= 252 \cdot 0,00060941 \approx 0,1536$$

1.2) Agora queremos

$$P[X \geq 5] = \sum_{x=5}^{10} \binom{10}{x} 0,35^x 0,65^{10-x} = \left[\binom{10}{5} 0,65^5 + \binom{10}{6} 0,65^4 0,35 + \dots + 0,35^5 \right] 0,35^5$$

$$= 0,00525 [252 \cdot 0,11603 + 210 \cdot 0,06248 + 120 \cdot 0,03364 + 45 \cdot 0,01812 + 10 \cdot 0,00975 + 0,00525]$$

$$= 0,00525 [29,23956 + 13,1208 + 4,0368 + 0,8154 + 0,0975 + 0,00525]$$

$$= 0,00525 [47,31531] = 0,2484.$$

1.3) Para $n=100$, temos

↖ *correção de continuidade*

$$P[X \geq 50] = P[49,5 \leq X \leq 100,5] = P\left[\frac{49,5-35}{4,7697} < \frac{X-\mu}{\sigma} < \frac{100,5-35}{4,7697}\right]$$

$$\approx P[3,04 < Z < 13,73] = \Phi(13,73) - \Phi(3,04) = 1 - 0,9988$$

$$= 0,0012.$$

- ② Temos que $X \sim \text{Pois}(\lambda)$, com $\lambda = E(X) = 3$.

2.1) $P[X=3] = \frac{e^{-3} 3^3}{3!} = \frac{e^{-3} \cdot 27}{6} = \frac{9}{2} \cdot e^{-3} = 4,5 \cdot e^{-3} = 0,224$

2.2) $P[X \geq 3] = 1 - P[X \leq 2] = 1 - P[X=0] - P[X=1] - P[X=2]$

$$= 1 - \left(\frac{e^{-3} \cdot 3^0}{0!} + \frac{e^{-3} \cdot 3^1}{1!} + \frac{e^{-3} \cdot 3^2}{2!} \right) = 1 - e^{-3} (1 + 3 + 4,5) = 1 - 8,5 \cdot e^{-3} = 0,5768$$

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3.1) Temos que $E[X] = \beta \Gamma(1 + \frac{1}{2}) = 100 \cdot \Gamma(1 + \frac{1}{0.5}) = 100 \cdot \Gamma(1+2)$
 $= 100 \cdot \Gamma(3) = 200.$

3.2) Note que

$$u = \left(\frac{t}{\beta}\right)^\alpha \Rightarrow du = \alpha \left(\frac{t}{\beta}\right)^{\alpha-1}$$

$$F(x) = \int_0^x \frac{\alpha}{\beta^\alpha} t^{\alpha-1} e^{-\left(\frac{t}{\beta}\right)^\alpha} dt \stackrel{\uparrow}{=} \int_0^{\left(\frac{x}{\beta}\right)^\alpha} e^{-u} du = -e^{-u} \Big|_0^{\left(\frac{x}{\beta}\right)^\alpha} = 1 - e^{-\left(\frac{x}{\beta}\right)^\alpha}.$$

Seja q tal que $F(q) = 0.5$. Logo,

$$0.5 = 1 - e^{-\left(\frac{q}{\beta}\right)^\alpha} \Rightarrow e^{-\left(\frac{q}{\beta}\right)^\alpha} = 0.5 \Rightarrow -\left(\frac{q}{\beta}\right)^\alpha = \log 0.5$$

$$\Rightarrow \left(\frac{q}{\beta}\right)^\alpha = (\log 2)^{1/\alpha} \Rightarrow q = \beta [\log 2]^{1/\alpha} = 100 (\log 2)^2 =$$

$$\Rightarrow q = 48.045.$$

4

Temos que $f(x,y) = K$ em $R = \{(x,y): -1 < x < 0, -x-1 < y < x+1, 0 < x < 1, x-1 < y < -x+1\}$. Assim,

$$1 = \int_{-1}^0 \int_{-x-1}^{x+1} K dy dx + \int_0^1 \int_{x-1}^{-x+1} K dy dx$$

$$= K \left[\int_{-1}^0 2x+2 dx + \int_0^1 -2x+2 dx \right] = K \left[(x^2+2x) \Big|_{-1}^0 + (-x^2+2x) \Big|_0^1 \right]$$

$$= K [-1+2 -1+2] = 2K \Rightarrow K = 1/2.$$

$$4.1) f_X(x) = \begin{cases} \int_{-x-1}^{x+1} \frac{1}{2} dy = x+1, & -1 < x < 0; \\ \int_{x-1}^{-x+1} \frac{1}{2} dy = -x+1, & 0 \leq x < 1 \end{cases}$$

$$f_Y(y) = \begin{cases} \int_{-y-1}^{y+1} \frac{1}{2} dx = y+1, & -1 < y < 0; \\ \int_{y-1}^{-y+1} \frac{1}{2} dx = -y+1, & 0 \leq y < 1. \end{cases}$$

4.2) Note que, tanto X quanto Y possuem fdp's marginais simétricas em torno de 0. Logo, devemos ter

$$E[X] = E[Y] = 0.$$

Agora

$$E[XY] = \int_{-1}^0 \int_{-x-1}^{x+1} xy \cdot \frac{1}{2} dy dx + \int_0^1 \int_{x-1}^{-x+1} xy \cdot \frac{1}{2} dy dx$$

$$= \int_{-1}^0 \frac{x}{2} \left[\frac{y^2}{2} \Big|_{-x-1}^{x+1} \right] dx + \int_0^1 \frac{x}{2} \left[\frac{y^2}{2} \Big|_{x-1}^{-x+1} \right] dx$$

$$= \int_{-1}^0 \frac{x}{4} [(x+1)^2 - (-x-1)^2] dx + \int_0^1 \frac{x}{4} [(-x+1)^2 - (-(-x+1))^2] dx$$

$$= 0.$$

Logo, $\text{Cov}(X, Y) = E[XY] - E[X] \cdot E[Y] = 0$.

4.3) Note que, mesmo tendo $\text{Cov}(X, Y) = 0$, temos que, por exemplo, para $0 < x, y < 1$

$$f_X(x) \cdot f_Y(y) = (-x+1)(-y+1) \neq \frac{1}{2} = f(x, y).$$

Logo, X e Y não são independentes.